Pete Millett Soundcard Interface Replica

Hardware Architecture & Design Baseline for M-Audio Solo

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System Parameter	Hardware Implementation Target
Active Analog Front-End	Dual OPA1656 CMOS Op-Amps (Ultra-low bias drift, 53 MHz GBW)
DC Offset Nulling	100 kΩ precision trim network (R_{25}) with onboard test pads (TP1/TP2)
Input Overvoltage Clamp	Back-to-back LND150 depletion MOSFETs + 1N4148W diodes (≤ 3 mA limit)
AC Coupling Threshold	0.47 μF 400 V Metallized PP Film ($f_{-3\text{dB}} \approx 3.3$ Hz, multi-pitch footprint)
Isolated Power Supply	XP Power JCA0305D03 (3 W, 5 V in, ±15 V out, 300 kHz, Faraday shield)
Power Input & Safety	USB-C (6-pin power) with 1812 SMD PTC fuse + SMCJ6.0A TVS clamp
Output Impedance Buffer	10 Ω native short-cable buffer (< 20 cm patch) + 36 Ω jumper pad
Chassis / Shield Ground	2-position Ground-Reference / Float toggle switch (SW2) on input BNC

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1 Executive Design Summary & Scope

This document establishes the electrical, mechanical, and PCB layout specifications for an optimized replica of the classic Pete Millett Soundcard Interface. The device serves as a precision analog front-end (AFE) attenuator, buffer, and galvanic isolation unit designed to protect computer audio interfaces while conducting laboratory-grade distortion (THD+N), frequency response, and noise floor measurements on audio equipment, including high-voltage vacuum tube amplifiers.

The design is customized specifically for tight bench integration with an M-Audio Solo audio interface situated at an operating distance of under 20 cm. While the original Pete Millett topology relied entirely on through-hole components, discrete analog metering, and an active signal generator driver, this revision achieves substantial bill-of-materials (BOM) cost reduction, board space optimization, and signal integrity improvements through:

1. **Architectural Pruning:** Total elimination of the legacy hardware generator output block and AD736JN-based true-RMS voltmeter. Modern measurement suites (REW, ARTA, Virtins Visual Analyser) handle excitation synthesis and calibrated digital signal tracking via software.
2. **Active Silicon Upgrades:** Transition to Texas Instruments OPA1656 Next-Gen CMOS operational amplifiers, eliminating DC offset errors caused by input bias currents flowing through the high-impedance (100 kΩ) divider.
3. **Solid-State Overload Clamping:** Substitution of high-noise 50 kΩ passive series protection resistors with depletion-mode MOSFET current limiters (Microchip LND150), reducing thermal Johnson noise voltage by a factor of four.
4. **Impedance Calibration for Short Transmission Lines:** Implementation of a dual-footprint daisy-chain isolation resistor network, yielding a native 10 Ω output impedance for maximum voltage transfer into the M-Audio Solo without parasitic oscillation.
5. **Modernized Power Infrastructure:** 6-pin power-only USB-C connector feeding an enhanced SMD crowbar overvoltage network and an XP Power JCA0305D03 metal-shielded isolated DC-DC converter.

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2 DC Power Supply & Input Protection Architecture

The power delivery network converts an unregulated +5 VDC supply into low-noise, fully isolated ±15 VDC analog split rails. Because the interface is powered via standard external USB power adapters or host PC ports, robust primary-side protection is mandatory to safeguard against overvoltage transients and polarity reversals.

2.1 Power Input Connector (USB-C 6-Pin)

Legacy full-sized USB-B and coaxial barrel jacks are deprecated in favor of a modern 6-pin USB-C receptacle (Power Only).

- **Connector Pinout:** Provides two V_{BUS} pins, two Ground pins, and two Configuration Channel (CC1/CC2) pins.
- **CC Logic Configuration:** To ensure universal enumeration with standard USB Type-C to Type-C smart power supplies (E-marked cables), CC1 and CC2 must each be terminated to ground through an independent $5.1\ \mathrm{k}\Omega \pm 1\%$ pulldown resistor.
- **Solderability:** The 6-pin layout provides wide (> 0.8 mm) pad spacing, permitting rapid, high-yield hand soldering compared to 24-pin high-speed data receptacles.

2.2 Active Crowbar Protection Subsystem

To prevent catastrophic failure should the user connect an off-spec power source (e.g., 9V, 12V, or 19V laptop chargers), an explicit two-stage crowbar protection circuit is positioned upstream of all active regulators.

a. **Series Overcurrent Limiter (F1):** Upgraded from the legacy radial leaded RUSBF110 to an 1812 surface-mount resettable PTC fuse ($I_{\mathrm{hold}} = 1.1$ A, $I_{\mathrm{trip}} = 2.2$ A, $V_{\mathrm{max}} \geq 16$ V). The 1812 package provides low internal resistance ($< 0.1\,\Omega$) and planar thermal dissipation.

b. **Shunt Clamping Diode (D9):** Replaces the axial 1.5KE9.1A with an SMD DO-214AB (SMC) Transient Voltage Suppressor (TVS), specifically the **SMCJ6.0A** (or SMCJ6.5A).

   - Reverse Standoff Voltage (V_{RWM}): 6.0 V.
   - Breakdown Voltage (V_{BR}): 6.67 V to 7.37 V.
   - Peak Pulse Power Dissipation: 1500 W.

   Engineering Justification: The original 9.1V breakdown diode allows dangerous voltage excursions before conduction, stressing the DC-DC input stage. A 6.0V standoff diode clamps immediately above nominal 5V, driving F1 into high-impedance latch-off within milliseconds.

2.3 Isolated DC-DC Converter (XP Power JCA0305D03)

- **Key Characteristics:** 3 W continuous rating, nominal +5 VDC input (4.5 V to 9.0 V operating window), fully regulated ±15 VDC dual output at ±100 mA.
- **Galvanic Isolation:** 1,500 VDC baseline isolation breaks ground continuity between the computer sound card and floating device-under-test (DUT) circuits.
- **EMI/RFI Containment:** Housed in an integrated nickel-coated copper metal enclosure. The conductive shell acts as an intrinsic Faraday cage, trapping high-frequency magnetic and electric fields.
- **Switching Profile:** Fixed switching frequency at 300 kHz. Because 300 kHz sits over an order of magnitude above the audible audio spectrum (20 Hz–20 kHz) and its sampling limits (192 kHz A/D Nyquist at 96 kHz), switching intermodulation artifacts are readily rejected.

2.4 Bulk Storage and High-Frequency Filtering Network

The legacy design specified 35V-rated radial electrolytic capacitors indiscriminately across both the 5V and 15V domains, consuming excessive board area. The revised component sizing is structured as follows:

- **Input Bulk Decoupling (C22):** Downrated from 35V to 10V or 16V (330 μF, low-ESR radial or aluminum SMD). Dimensions reduce from a diameter of 10 mm to 6.3 mm.
- **High-Frequency Input Snubber:** An 0805 0.1 μF X7R ceramic capacitor is routed directly across pins 22/23 ($V_{\mathrm{IN}+}$) and pins 2/3 ($V_{\mathrm{IN}-}$) of the JCA module. Electrolytic bulk capacitors exhibit significant inductive reactance at 300 kHz; the local ceramic shunt captures inductive current spikes before they radiate across the input plane.
- **Output Rail Filtering (C7, C8, C13, C18):** Downrated from 35V to 25V (330 μF low-ESR). Sized for reliable headroom above the continuous ± 15 VDC rail potential while minimizing footprint height.

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3 Analog Front-End (AFE) Signal Conditioning

The input stage accepts single-ended or balanced voltages ranging from line-level audio ($< 1\,\mathrm{V_{RMS}}$) up to high-voltage tube amplifier plate swings ($> 100\,\mathrm{V_{RMS}}$).

3.1 High-Voltage DC Blocking Capacitors (C2, C3)

DC blocking capacitors isolate the sensitive solid-state buffer from the high DC anode potentials encountered in vacuum tube circuits.

- **Capacitance Optimization:** Pete Millett originally specified 1.0 μF 400V polyester dielectric blocks to extend sub-audible measurement performance down to 1.5 Hz. For standard audio analysis (20 Hz–20 kHz), the input coupling values are downsized to 0.47 μF (400VDC working rating).
- **Cut-off Calculation:** Across the 100 kΩ system input impedance, the high-pass -3 dB corner frequency is given by:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 10^5 \times (0.47 \times 10^{-6})} \approx 3.38\,\mathrm{Hz} \tag{1}$$

  At 20 Hz, the magnitude attenuation is below 0.06 dB, and phase shift is negligible.
- **Dielectric Selection:** Metallized Polypropylene (PP) Film (e.g., WIMA MKP4 or Panasonic ECQ-E). High-voltage Class II SMD ceramic capacitors (X7R, X5R) are strictly prohibited due to severe Voltage Coefficient of Capacitance (VCC) non-linearities, which inject harmonic distortion under high-voltage AC swings.
- **CAD Footprint Provisioning:** The PCB layout incorporates multi-pitch radial through-hole pads with concentric drill points spaced at 5.0 mm, 7.5 mm, 10.0 mm, and 15.0 mm, ensuring compatibility with various film capacitor form factors.

3.2 Active Protection Clamping & Low-Noise Current Limiting

When an overvoltage transient enters the front-end, high currents must be prevented from damaging the op-amp input stages.

1. **Depletion-Mode MOSFET Replacement (R2, R16):** The original circuit used 49.9 kΩ, 2 W carbon/metal film resistors to limit fault current into the protection diodes. These resistors introduce significant thermal Johnson noise:

$$v_n = \sqrt{4k_B T R \Delta f} \approx 28.7\,\text{nV}/\sqrt{\text{Hz}} \quad (\text{at } 300\,\text{K}) \tag{2}$$

   This noise creates an elevated noise floor, obscuring micro-detail in low-distortion audio testing.

   Modern Implementation: The fixed 50 kΩ passive resistors are replaced with a back-to-back series pair of **Microchip LND150** depletion-mode N-channel MOSFETs (SOT-89 or TO-92).

   - *Topology:* Source pins are tied together; the Gates are tied directly to the common Source center node.
   - *Normal Audio Conduction:* Under normal operating voltages ($|V| < \pm 15$V), both channels remain fully enhanced, exhibiting an incremental resistance of only ~ 1.5 kΩ to 3 kΩ. Thermal noise voltage drops to $\sim 7.0\,\text{nV}/\sqrt{\text{Hz}}$—a four-fold (12 dB) improvement in front-end SNR.
   - *Overvoltage Limiting:* When a high-voltage spike occurs, the differential gate-source voltage pinches off the conduction channel, saturating the pass current to a fixed 1.5 mA to 3.0 mA.

2. **Input Clamping Diodes:** The fault current is sunk into the power rails via high-speed silicon switching diodes. **1N4148W** (SOD-123 surface mount) parts are selected. Standard power rectifiers or Schottky diodes are avoided because their high junction capacitance (> 30 pF) forms an unintended low-pass filter with the attenuator ladder, rolling off high-frequency phase response. The 1N4148W maintains a junction capacitance under 4 pF.

3.3 Precision Attenuator Ladder & Rotary Switch Interface

Attenuation is handled through a precision 4-range voltage divider network (R4 through R7) configured to yield stepped attenuation scales (0 dB, −20 dB, −40 dB, −60 dB).

- **Switch Topology Simplification:** With the elimination of the dual-rail analog metering circuit, the original multi-wafer 2-pole switch is simplified to a single-pole four-throw (1P4T) configuration.
- **Component Selection:** Standard Alpha SR2611F 1-Pole, 12-Position rotary switch. The included adjustable metal mechanical stop ring is locked at hole position 4, providing an exact, positive-stop 4-position action while avoiding custom low-volume multi-deck switch costs.
- **Wiring Harness:** The switch is wired off-board using 5 active conductors (1 common input lead and 4 range taps). These conductors are bundled and tightly twisted to maintain high common-mode magnetic rejection.

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4 Analog Buffer, Instrumentation Amplifier & DC Nulling

4.1 Operational Amplifier Selection: Texas Instruments OPA1656IDR

The dual operational amplifier (U4A/U4B) serves as the high-impedance buffer stage. Precision measurements demand that the op-amp introduce negligible loading error and zero DC drift across the attenuator.

Parameter	**NE5532 (Legacy Bipolar)**	**OPA2134 (Legacy JFET)**	**OPA1656 (Selected CMOS)**
Input Bias Current (I_B)	200 nA (typ), 800 nA (max)	5 pA (typ), 100 pA (max)	**10** pA (typ), **25** pA (max)
Voltage Noise Density (e_n)	5.0 nV/$\sqrt{\text{Hz}}$	8.0 nV/$\sqrt{\text{Hz}}$	**2.9** nV/$\sqrt{\text{Hz}}$
Gain-Bandwidth Product (GBW)	10 MHz	8 MHz	**53** MHz
Slew Rate (SR)	9 V/μs	20 V/μs	**24** V/μs
THD+N (1 kHz, 2 kΩ)	0.0005%	0.00008%	**0.000029**% (−131 dB)
Input Offset Voltage Drift	2 μV/°C	2 μV/°C	**0.5** μV/°C
Package / Availability	DIP-8 / SOIC-8	DIP-8 / SOIC-8	SOIC-8 (SMD Surface Mount)

Table 1: Parametric comparison demonstrating the operational advantages of the OPA1656 CMOS amplifier.

DC Offset Analysis: Across the worst-case 100 kΩ input resistance, a traditional bipolar op-amp (NE5532, $I_B \approx 72$ nA) develops an uncalibrated DC offset:

$$V_{\text{offset}} = I_B \times R_{\text{in}} = 72 \times 10^{-9}\,\text{A} \times 100 \times 10^{3}\,\Omega = 7.2\,\text{mV} \tag{3}$$

In contrast, the OPA1656 generates:

$$V_{\text{offset}} = 10 \times 10^{-12}\,\text{A} \times 100 \times 10^{3}\,\Omega = 0.001\,\text{mV} \quad (1.0\,\mu\text{V}) \tag{4}$$

This negligible DC offset preserves high DC stability without loading the calibration system.

4.2 Differential Subtraction & Instrumentation Topology

- **Dual Buffer Topology:** U4A buffers the positive input path (V_{INP}), while U4B buffers the input reference/shield path (V_{INN}).
- **Common-Mode Hum Cancellation:** Both channels feed into a downstream high-CMRR difference amplifier (Texas Instruments INA134 or equivalent difference stage U7). Any electromagnetic noise, mains hum (60 Hz or 50 Hz), or radio-frequency pickup coupled equally to the BNC center pin and outer shield appears identically at the inputs of U4A and U4B and is canceled out by U7.
- **Chassis Ground-Reference Switch (SW2):** SW2 allows the user to switch the input BNC shield between direct circuit ground and a floating configuration. When ground loops between the DUT, mains ground, and host soundcard corrupt the noise floor, opening SW2 breaks the loop while U4B maintains proper signal-path referencing.

4.3 DC Nulling Calibration Stage (R25, TP1, TP2)

To achieve laboratory-grade measurement accuracy, remaining input offset voltages can be nulled using the on-board calibration trim:

- **Components:** A 100 kΩ multi-turn cermet trimmer (R25) connected to the offset trim pins, paired with two surface-mount loop test points (TP1 and TP2).
- **Calibration Procedure:**
  1. Terminate the input BNC with a 50 Ω or 75 Ω shorting cap; leave the interface powered on for 15 minutes to reach thermal equilibrium.
  2. Connect a $4\frac{1}{2}$-digit or $6\frac{1}{2}$-digit digital multimeter (DMM) set to the DC millivolt range across TP1 and TP2.
  3. Adjust R25 until the measured potential is $0.00\,\text{mV} \pm 10\,\mu\text{V}$.

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5 Oscillation Prevention, Output Stability & Cable Interface

The Texas Instruments OPA1656 is a wide-bandwidth operational amplifier ($f_{\text{unity}} = 53$ MHz). At these frequencies, parasitic board capacitances and load cable capacitances can erode amplifier phase margin, resulting in high-frequency ringing or sustained oscillation.

5.1 Active Layout Stability Rules

a. Direct Rail Decoupling: High-frequency surface-mount ceramic bypass capacitors (0.1 μF X7R, 0805) are placed directly against Pin 8 ($V+$) and Pin 4 ($V-$). Power traces route through the capacitor solder pad prior to entering the IC lead frame. A 10 μF tantalum or low-ESR electrolytic capacitor is placed within 10 mm upstream on each rail.

b. Feedback Phase Compensation ($C_{\mathbf{fb}}$): An 0805 capacitor footprint is routed in parallel with the main feedback resistor (R_{fb}). This footprint is populated with a 10 pF to 22 pF C0G/NP0 dielectric ceramic capacitor. This small feedback capacitance introduces a high-frequency zero in the loop transmission, providing phase lead to stabilize the internal pole against parasitic input capacitance.

5.2 Output Isolation Resistor ($R_{\mathbf{iso}}$) Analysis for M-Audio Solo

Driving capacitive loads directly from low-impedance op-amp outputs degrades phase margin according to:

$$f_{\text{pole}} = \frac{1}{2\pi R_o C_{\text{load}}} \tag{5}$$

Pete Millett's baseline design specified a conservative 47 Ω series output isolation resistor, engineered to drive up to 5 meters of audio cabling ($C_{\text{cable}} \approx 500$ pF to 1000 pF).

5.2.1 Short-Cable Bench Condition (< 20 cm)

For the dedicated M-Audio Solo installation:

- Cable Length: ≤ 20 cm shielded unbalanced patch lead.
- Total Parasitic Capacitance: $C_{\text{cable}} \leq 25$ pF to 30 pF.
- Instrument/Line Input Impedance of M-Audio Solo: ~ 10 kΩ (Line) / ~ 1 MΩ (Instrument).

Evaluation of $10\,\Omega$ *vs.* $47\,\Omega$*:* With only 30 pF of load capacitance, the secondary pole introduced by a $10\,\Omega$ resistor lies above 500 MHz, well outside the amplifier unity-gain crossover. Dropping R_{iso} to $10\,\Omega$ minimizes voltage attenuation across the interface-to-soundcard junction, preserving absolute calibration accuracy for precision voltmeters and software analyzers.

5.2.2 Mechanical Switch Prohibition & Daisy-Chain Footprint Implementation

Routing front-panel mechanical switches to toggle between $10\,\Omega$ and $47\,\Omega$ is prohibited. Running unshielded traces to a toggle switch adds several picofarads of stray capacitance and acts as an RF antenna, creating the exact instability the resistor is designed to prevent.

PCB Design Solution (Series Daisy-Chain Network): A series dual-resistor footprint with a bridgeable jumper pad is placed directly adjacent to the op-amp output pin:

$$\boxed{\text{OPA1656 Out}} \longrightarrow [\ 10\,\Omega,\ 0805\ ] \longrightarrow \textbf{Node A} \longrightarrow [\ 36\,\Omega,\ 0805\ ] \longrightarrow \boxed{\text{Output Jack}}$$

$$\Big|\qquad\qquad\Big|$$

$$\text{— [SMT Solder Jumper] —}$$

- **Bench Configuration (M-Audio Solo):** Both the $10\,\Omega$ and $36\,\Omega$ resistors are populated. A solder blob or low-resistance zero-ohm jumper is bridged across the $36\,\Omega$ resistor pad, shunting it out. The circuit operates natively at $10\,\Omega$ nominal impedance.
- **Field Configuration (Long Cable Drive):** Should the interface be repurposed to drive long cables (> 3 m), the jumper bridge is opened with a soldering iron or scalpel. This activates the $36\,\Omega$ element in series with the $10\,\Omega$ resistor, establishing a $46\,\Omega$ protection buffer without requiring PCB modifications.

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6 Balanced Output Line Driver Provisioning (DRV135UA)

Although the current M-Audio Solo hardware uses an unbalanced 1/4-inch input, the PCB layout provisions for a future balanced XLR output to support higher-end audio analyzers.

- **Silicon Optimization:** The legacy DIP-8 DRV134PA is superseded by the surface-mount **Texas Instruments DRV135UA** (SOIC-8).

  - THD+N: 0.0005% at 1 kHz.
  - Slew Rate: $15\,\mathrm{V}/\mu\mathrm{s}$.
  - Package: Compact SOIC-8, reducing board real estate by over 60%.

- **Unpopulated Footprint Strategy:** The DRV135UA, its bypass capacitors, and the chassis-mount 3-pin XLR connector footprints are etched on the PCB but left unpopulated. The single-ended output from the INA134 difference stage is routed directly to the primary 1/4-inch phone jack. When an audio interface upgrade occurs, the DRV135UA and XLR jack can be populated to activate the differential balanced output.

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7 System Chassis Wiring & Mechanical Integration

To ensure high-frequency stability with the 53 MHz OPA1656 and preserve high CMRR, off-board panel controls must follow strict wiring practices:

Circuit Path	**Conductor Type**	**Routing Specification**
Main DC Power (V_{BUS}, GND)	18 AWG Multi-strand Copper	Direct route from USB-C breakout to PCB power terminal; minimizes DC IR drop.
Analog Audio Inputs & Outputs	24 AWG or 26 AWG Shielded Coax	Miniature RG-178 or high-flex shielded audio cable; drain shield tied at PCB star point.
Rotary Attenuator Controls	24 AWG Discrete Twisted Harness	Tightly twisted bundle (≥ 3 twists per cm) to ensure external magnetic fields cancel.
BNC Shield Ground Switch (SW2)	22 AWG Solid Copper	Shortest direct line from BNC outer shell to toggle switch and star grounding lug.

Table 2: Interconnect and chassis wiring schedule.

7.1 Chassis Construction and RF Shielding

The complete assembly must be enclosed in an extruded aluminum enclosure (e.g., Hammond 1455 series). The aluminum chassis connects to signal ground at a single star point adjacent to the power input. All anodized layers must be cleared with a star washer to ensure low contact resistance between panels and chassis ground.

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8 Consolidated Hardware Bill of Materials (BOM)

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Designator	Function	Original Part	Selected Part	Package / Notes
U1	Isolated DC-DC	JCA0305D03	JCA0305D03	Metal Case, 3W, 5V to ±15V
U4A, U4B	Dual Input Buffer	NE5532 / OPA2134	**OPA1656IDR**	SOIC-8, CMOS, 53MHz, 2.9nV/$\sqrt{\text{Hz}}$
U7	Difference Amp	INA134PA	**INA134UA**	SOIC-8, Precision Difference Amp
U8 (Opt.)	Balanced Line Driver	DRV134PA	**DRV135UA**	SOIC-8 (Footprint unpopulated)
F1	Resettable Fuse	RUSBF110 (Radial)	**SMD1812-110**	1812 SMT, 1.1A Hold, 2.2A Trip
D9	TVS Clamp Diode	1.5KE9.1A (Axial)	**SMCJ6.0A**	DO-214AB (SMC), 6.0V V_{RWM}
D1–D8	Input Clamping	1N4148 (DO-35)	**1N4148W**	SOD-123, Low Capacitance (< 4 pF)
Q1–Q4	Overload Limiter	49.9 kΩ 2W Res.	**LND150N8-G**	SOT-89 Depletion MOSFET

Table 3: Active Silicon, Power & Protection Bill of Materials.

9 Summary & Next Steps

This engineering specification establishes a low-noise, cost-effective test front-end matched to the M-Audio Solo. By removing legacy measurement circuits, upgrading to modern CMOS and depletion-mode solid-state devices, and implementing a configurable 10 Ω output stage, the interface delivers professional measurement performance in a compact bench footprint.

The immediate follow-on engineering tasks are:

1. Generate the complete schematic netlist in KiCad including all unified power, protection, attenuator, and buffer sheets.
2. Finalize the mechanical drilling template for the Hammond extruded enclosure front/rear panels.
3. Execute PCB routing following the strict 2-layer star-grounding and high-frequency decoupling layout guidelines outlined in Section 5.

Designator	Function	Original Part	Selected Part	Package / Notes
C2, C3	DC Blocking Caps	1.0 μF 400V Film	0.47 μ**F 400V PP**	WIMA MKP4, Multi-pitch PTH
C22	Input Bulk Cap	330 μF 35V	330 μ**F 16V**	Low-ESR Radial Electrolytic
C7,8,13,18	Output Rail Bulk	330 μF 35V	330 μ**F 25V**	Low-ESR Radial Electrolytic
$C_{\text{in_hf}}$	Switching Snubber	None	0.1 μ**F X7R**	0805, 50V Ceramic, Parallel to C22
$C_{\text{fb1,2}}$	Compensation	None	15 **pF C0G**	0805 Ceramic, Across feedback
R_{iso1}	Output Isolation A	47 Ω Carbon Film	10 Ω **1%**	0805 Thin Film
R_{iso2}	Output Isolation B	None	36 Ω **1%**	0805 Thin Film, with SMT Jumper
SW1	Range Switch	2-Pole 6-Pos Rotary	**Alpha SR2611F**	1-Pole 12-Pos, stopped at Pos 4
SW2	Shield Reference	SPDT Leaded Toggle	**SPDT Sub-Mini**	Panel Mount, Break-before-make
J4	DC Power In	USB-B / Barrel	**USB-C 6-Pin**	Surface Mount, Power-only with 5.1 kΩ

Table 4: Passive Components & Electro-Mechanical Hardware Bill of Materials.